

HALIFAX DOCKYARD, NS, Canada

WMO: 713280

Lat: 44.660N

Lon: 63.580W

Elev: 4

StdP: 101.28

Time Zone: -4.00 (NAA)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.6	-11.6	-18.9	0.7	-12.0	-17.2	0.8	-9.7	15.6	0.5	12.4	-1.0	4.3	270	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	27.2	19.2	25.5	18.5	24.0	17.6	21.1	24.9	20.1	23.4	19.3	22.2	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.6	22.4	19.0	13.8	21.6	18.1	13.1	20.8	61.0	24.9	57.8	23.3	55.1	22.3	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	9.6	8.4	-16.7	30.9	1.7	1.9	-17.9	32.2	-19.0	33.3	-19.9	34.4	-21.2	35.7		
(5)			-17.2	22.7	1.6	0.8	-18.3	23.3	-19.2	23.7	-20.1	24.2	-21.3	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-2.2	-2.3	0.4	5.4	9.9	14.6	19.3	19.7	16.6	11.6	6.1	0.8
	DBStd	8.61	5.17	4.55	4.09	3.17	3.06	3.09	2.56	2.16	2.86	3.58	4.01	4.82
	HDD10.0	1640	379	344	298	143	39	2	0	0	0	24	127	284
	HDD18.3	3753	637	578	555	389	262	118	18	11	66	211	366	542
	CDD10.0	1052	0	0	1	3	36	139	289	301	198	73	11	1
	CDD18.3	124	0	0	0	0	1	6	49	54	14	1	0	0
	CDH23.3	483	0	0	2	1	12	43	222	170	31	3	0	0
	CDH26.7	78	0	0	0	0	2	9	45	21	2	0	0	0
	WSAvg	3.8	3.9	4.2	4.3	4.2	3.9	3.6	2.9	3.0	3.3	3.8	4.1	4.1
	PrecAvg	1414	132	114	123	110	111	103	101	96	97	128	139	151
(15) Precipitation	PrecMax	1795	271	229	243	208	214	349	229	320	225	256	258	253
	PrecMin	983	42	34	45	19	26	32	13	25	27	32	38	67
	PrecStd	198	47	47	46	42	45	55	48	61	46	52	55	51
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	10.5	14.5	17.7	24.8	27.5	30.6	29.0	26.4	22.7	17.1	13.9
		MCWB	11.0	9.2	9.3	10.5	15.1	17.5	20.9	20.7	19.4	16.3	15.3	12.6
	2%	DB	9.1	7.7	9.7	14.0	19.8	24.1	27.6	26.7	23.9	20.0	15.5	11.6
		MCWB	8.0	6.3	6.1	8.1	12.5	16.2	19.6	19.3	18.3	16.2	13.7	10.6
	5%	DB	7.1	5.6	7.6	11.9	16.9	21.9	25.8	25.2	22.3	18.1	13.8	9.6
		MCWB	5.9	4.3	4.9	8.0	11.1	15.1	18.7	18.3	17.5	15.5	12.2	8.6
	10%	DB	5.1	3.9	5.8	10.1	14.7	19.8	23.9	23.7	20.8	16.7	12.3	7.8
		MCWB	3.8	2.5	3.4	6.9	10.1	14.5	17.8	17.8	17.3	14.4	10.7	6.7
	0.4%	WB	11.0	9.7	10.6	12.9	17.1	19.7	22.4	22.2	21.4	19.4	15.9	13.0
		MCDB	11.8	10.2	13.5	15.4	22.6	24.5	26.7	26.2	24.1	21.0	16.8	13.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	8.3	6.6	7.2	10.6	14.5	17.9	21.2	21.0	20.0	17.4	14.2	10.5
		MCDB	9.1	7.3	8.8	12.1	17.5	22.2	25.5	24.4	22.3	18.9	15.2	11.3
	5%	WB	6.1	4.6	5.4	8.8	12.5	16.6	20.1	20.1	19.0	16.0	12.5	8.5
		MCDB	7.0	5.5	7.1	10.7	15.4	19.7	24.0	23.0	21.1	17.5	13.5	9.4
	10%	WB	3.9	2.8	3.7	7.4	10.9	15.4	19.2	19.3	18.0	14.5	10.9	6.6
		MCDB	4.9	3.7	5.5	9.3	13.6	18.4	22.4	21.9	19.8	16.5	12.1	7.5
	5% DB	MDBR	7.0	6.6	6.6	6.7	7.3	7.2	7.5	7.2	6.8	6.0	5.9	6.3
		MCDBR	9.3	8.6	9.1	10.1	11.8	11.9	10.7	9.5	8.6	7.6	7.6	8.6
		MCWBR	9.4	8.3	7.2	6.5	6.4	5.5	4.5	4.1	4.7	5.3	6.9	8.8
		MCWBR	9.3	8.7	8.7	8.5	10.3	9.7	9.0	7.8	7.1	6.9	7.4	8.2
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.7	8.6	8.0	7.0	6.9	5.5	5.1	4.3	5.2	6.1	7.6	8.9
(40) Clear-Sky Solar Irradiance	taub		0.304	0.297	0.325	0.350	0.371	0.401	0.417	0.398	0.355	0.333	0.317	0.297
	taud		2.464	2.517	2.470	2.443	2.430	2.373	2.350	2.400	2.492	2.551	2.549	2.533
	Ebn at Noon		818	902	913	913	899	869	850	855	869	846	801	795
	Edh at Noon		71	80	96	107	112	119	120	111	93	77	65	60
(44) All-Sky Solar Radiation	RadAvg		1.43	2.33	3.52	4.30	4.95	5.38	5.36	5.00	4.06	2.63	1.51	1.13
	RadStd		0.11	0.17	0.24	0.36	0.63	0.50	0.39	0.35	0.27	0.27	0.16	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	+0.36	+0.50	N/A	N/A	+50

Nomenclature: See separate page